① (a) $\sum_{n=1}^{\infty} \sqrt{\frac{n}{2(n+1)}}$ $\quad a_n = \sqrt{\frac{n}{2(n+1)}} \Rightarrow \lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{\sqrt{2}} \sqrt{\frac{1}{1+1/n}} = 1/\sqrt{2}$

$\neq 0$ Hence $\sum a_n$ is div.

(b) $\sum_{n=1}^{\infty} n \sin \frac{1}{n}$ $\quad a_n = n \sin \frac{1}{n} \Rightarrow \lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{\sin 1/n}{1/n} = 1 \neq 0$

Hence div.

(c) $\sum_{n=1}^{\infty} \sin \frac{n\pi}{3} = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + 0 - \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} + 0 + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + 0 + \cdots$

sequence of partial sums $S_n = \{ \underset{1}{\frac{\sqrt{3}}{2}}, \underset{2}{\sqrt{3}}, \underset{3}{\sqrt{3}}, \underset{4}{\frac{\sqrt{3}}{2}}, \underset{5}{0}, \underset{6}{0}, \underset{7}{\frac{\sqrt{3}}{2}}, \underset{8}{\sqrt{3}} \cdots \}$

$\lim_{n\to\infty} S_{6n-5} = \sqrt{3}/2$ & $\lim_{n\to\infty} S_{6n-4} = \sqrt{3}$ limit not unique

Hence (not conv.) osci. finite

② (a) $a_n = \frac{2n-1}{n(n+1)(n+2)}$ $\quad$ Let $b_n = \frac{1}{n^2}$ then

$\frac{a_n}{b_n} = \frac{(2n-1)n^2}{n(n+1)(n+2)} = \left(\frac{n}{n+1}\right) \cdot \left(\frac{2n-1}{n+2}\right)$ $\quad \therefore \lim_{n\to\infty} \frac{a_n}{b_n} = 2 \neq 0$ & finite

So, by limit comparison test, $\sum a_n$ & $\sum b_n$ converge & diverge together. $\because \sum b_n$ is cgt $\Rightarrow \sum a_n$ is cgt.

(b) $a_n = \sqrt{n+1} - \sqrt{n} = \frac{\sqrt{n+1} - \sqrt{n}}{1} \times \frac{\sqrt{n+1} + \sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}$

Let $b_n = \frac{1}{\sqrt{n}}$ then $\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\sqrt{n}}{\sqrt{n+1} + \sqrt{n}} = \lim_{n\to\infty} \frac{1}{\sqrt{1+\frac{1}{n}} + 1} = 1 \neq 0$ & finite

$\therefore$ by limit comparison test $\because \sum b_n$ is div $\Rightarrow \sum a_n$ is also div.

(c) $a_n = \sin \frac{1}{n^2}$ $\quad$ Take $b_n = 1/n^2$ then

$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\sin(1/n^2)}{1/n^2} = 1 \neq 0$ & finite $\quad \lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$

By limit comparison test $\because \sum b_n$ is cgt $\Rightarrow \sum a_n$ is also cgt.

③ (a) $a_n = \frac{n^{n^2}}{(n+1)^{n^2}}$

$= \left[\frac{n}{n+1}\right]^{n^2}$

Now $(a_n)^{1/n} = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(1+\frac{1}{n}\right)^n}$

$\lim_{n\to\infty} (a_n)^{1/n} = \lim_{n\to\infty} \frac{1}{\left(1+\frac{1}{n}\right)^n} = 1/e < 1$

By cauchy's root test, $\sum a_n$ is cgt.

(b) $a_n = [n^{1/n}+1]^n \Rightarrow \lim_{n\to\infty} (a_n)^{1/n} = \lim_{n\to\infty} n^{1/n}+1 = 1+1 = 2 > 0$

$\because \lim_{n\to\infty} n^{1/n} = 1$ & $p^{1/n}$ for p fixed.

Hence Σa_n is div. by cauchy root test

(c) $a_n = \dfrac{2^{n+1}}{3^n+1}$, $a_{n+1} = \dfrac{2^n}{3^{n+1}+1} \Rightarrow \dfrac{a_{n+1}}{a_n} = 2 \cdot \dfrac{3^n+1}{3^{n+1}+1} = 2 \cdot \dfrac{1+1/3^n}{3+1/3^n}$

$\lim_{n\to\infty} \dfrac{a_{n+1}}{a_n} = \dfrac{2}{3} < 1$ Hence Σa_n is cgt by Ratio test.

(d) $a_n = \dfrac{n^n}{n!}$, $a_{n+1} = \dfrac{(n+1)^{n+1}}{(n+1)!} \Rightarrow \dfrac{a_{n+1}}{a_n} = \dfrac{(n+1)^{n+1}}{n^n} \cdot \dfrac{n!}{(n+1)!} = \dfrac{(n+1)^n}{n^n}$

$\lim_{n\to\infty} \dfrac{a_{n+1}}{a_n} = \lim_{n\to\infty} \left(1+\dfrac{1}{n}\right)^n = e > 1$ Hence Σa_n is div. by Ratio test.

<u>NOTE:</u> $\lim_{n\to\infty} [1+f(n)]^{g(n)} = e^{f(n)\cdot g(n)}$

(4) (a) $a_n = (-1)^n \dfrac{\cos n\alpha}{n^3} \Rightarrow |a_n| = \dfrac{|\cos n\alpha|}{n^3} \le \dfrac{1}{n^3}$

& $\Sigma \dfrac{1}{n^3}$ is cgt by p-series test $\therefore \Sigma |a_n|$ is cgt by comparison test $\Rightarrow \Sigma a_n$ is absolutely cgt $\Rightarrow \Sigma a_n$ is cgt also.

(b) $\because \log x$ is an increasing f^n & $n+2 > n+1$, we have

$\log(n+2) > \log(n+1) \Rightarrow \dfrac{1}{\log(n+2)} < \dfrac{1}{\log(n+1)} \Rightarrow a_{n+1} < a_n \quad \forall n$

i.e. a_n is monotonically decreasing seq. and

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \dfrac{1}{\log(n+1)} = 0$. Thus, by Leibniz test $\Sigma(-1)^n a_n$ is cgt.

Further $\log(n+1) < n+1 \ \forall n \Rightarrow \dfrac{1}{\log(n+1)} > \dfrac{1}{n+1}$ & $\Sigma \dfrac{1}{n+1}$ is div because $\Sigma \dfrac{1}{n}$ is div. Hence by comparison test, $\Sigma \dfrac{1}{\log(n+1)}$ is div. which means ~~$\Sigma|a_n|$ is div~~ $\Sigma \left| \dfrac{(-1)^{n+1}}{\log(n+1)} \right|$ is div.

This implies

$\Sigma \dfrac{(-1)^{n+1}}{\log(n+1)}$ is conditionally cgt.